

Primary Hepatic Cancer and Liver Cirrhosis Autopsy Study Covering Fifty Years

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Summary

Autopsy reports from 1931 to 1980 were used to study the incidence of liver cirrhosis (LC) and the association between LC and hepatocellular carcinoma (HCC) in our area (Wuppertal, Germany). An increase in LC and in LC with HCC has occurred since World War II, with HCC being most frequently associated with postnecrotic cirrhosis. The prevalence of HCC in men with LC was highest (13.5%) in 1966-1970, whereas the prevalence of HCC with LC in women rose abruptly to a peak (11.8%) during the last 5 years of the study. Possible etiological factors for the association between LC and HCC are discussed.

Key-Words: Hepatic Cancer — Liver Cirrhosis — Autopsy reports — Incidence of liver cirrhosis

Introduction

Many recent national and international studies indicate that the incidence rates of liver cirrhosis (LC) and primary hepatocellular carcinoma (HCC) are on the increase (1, 3, 4, 8, 12, 25, 32, 45, 46). Furthermore, there appears to be a close association between these diseases. Weiss and Hanak (52) for example, calculated, that a malignant change occurs in the liver in 30% of patients with LC, while others have found that between 52% and 90% of HCC cases are seen in patients with LC (22, 23, 28, 29, 34, 39). A causal connection between HCC and LC has, in fact, been suspected ever since Eggel studied these diseases in 1901 (16). Unfortunately, the etiological factors responsible for these diseases are still incompletely known.

On way of obtaining information about the causal factors of diseases is by studying the incidence of disease at various places and various times. Several previous studies have reported geographical differences in LC and HCC; both diseases are more frequent in Africa and Southeast Asia than in Europe and North America (19, 20, 29, 31, 33, 38, 41, 44). Differences in the incidence rates of LC and HCC in men and women are also common; both diseases tend to occur more often in men than in women (11, 13, 21, 25, 26, 33, 47). Less is known, however, about temporal changes in LC and HCC. The present study was carried out to study temporal changes in the incidence of LC and HCC in men and women, using autopsy material collected over 50 years.

Materials and Methods

This study is based on 26,864 autopsy reports made between January 1, 1931 and December 31, 1980 at the Institute of Pathology at the Municipal Hospital of Wuppertal. The data of 1992 autopsies had to be omitted, however, because they were not complete enough for our purposes, mainly due to war-time problems. The remaining data comprised 13,154 adult males, 8848 adult females, one hermaphrodite and 2869 children up to the age of 15 years. We reclassified liver cirrhosis (LC) from all autopsy reports according to the guidelines of the International Committee for Classification and Nomenclature of Liver Cirrhosis of the 5th Panamerican Congress of Gastroenterology 1956 in Havana (4, 45). We used the histological classification of liver tumors given in the original autopsy reports, however, because original slides of some early cases could not be obtained, and because a reclassification only on the basis of autopsy documents could not be done with sufficient accuracy.

Results

Table 1 shows the incidence rate of LC in adults. LC was found on autopsy in 1014 males and 571 females. Primary LC was the most frequent type in both sexes. Primary HCC was seen together with LC in 9.8% of men and 3.3% of women, with postnecrotic LC being the type most often associated with HCC in both sexes (17.5% of males and 4.6% of females). The incidence rate of HCC together with portal LC was also greater in males (6.8%) than in females (2.6%).

Table 2 shows the data for adults divided into 5-year periods. In general, there was an increase in LC and an increasing association of LC and HCC over the 50-year period. The incidence of LC and LC with HCC was relatively high during 1931 and 1935 as compared with the ten following years. After World War II, both LC and LC with HCC tended to increase gradually for 25 years. No further overall increase in LC occurred during the last 5 years, whereas the percentage of LC with HCC has continued to rise. It is worthy of note, however, that a marked increase in LC with HCC occurred in women during the last 5 years of the study.

LC was present in 11 of the children (0.38%). No further analysis of data for children is presented in this report.

Table 1 Incidence of liver cirrhosis (LC) and hepatocellular carcinoma (HCC) in adults based on 22,002² autopsy reports from January 1, 1931 to December 31, 1980

		Total	Morphological classification of LC ¹			
			Portal	Postnecrotic	Biliary	Unclassified
Both sexes						
Total LC	(n)	1585	978	409	102	96
	(%)	7.2	4.4	1.9	0.5	0.4
LC with HCC	(n)	118	52	55	2	9
	(%)	7.5	5.3	13.5	2.0	9.4
Males (13, 154)		(n)				
Total LC	(n)	1014	630	280	37	67
	(%)	7.7	4.8	2.1	0.3	0.5
LC with HCC	(n)	99	43	49	0	7
	(%)	9.8	6.8	17.5	0	6.9
Females (8, 848)		(n)				
Total LC	(n)	571	348	129	65	29
	(%)	6.5	3.9	1.5	0.7	0.3
LC with HCC	(n)	19	9	6	2	2
	(%)	3.3	2.6	4.6	0.4	0.4

¹ According to guidelines of the International Committee for Classification and Nomenclature of Liver Cirrhosis of the 5th Panamerican Congress of Gastroenterology, 1956 (4, 45)

² One hermaphrodite was excluded from 22,003 adults

Discussion

Our findings agree with previous reports showing an association between LC and HCC; we found that 7.5% (1 out of 13) of adults with LC also had HCC at the time of death. Our results accord with the outcome of most other studies carried out in the West on LC and HCC (2, 4, 7, 9, 43, 53). It is worthy of note, however, that some older studies have estimated the degree of association between LC and HCC to be lower than 5%, whereas most recent studies show the association to be higher. Thus, the increasing trend in recent years for LC to be found together with HCC may account for some discrepancies between studies. Nevertheless, the findings support the notion that LC may be the most important etiological factor for the formation of HCC (30).

Our findings in agreement with reports that HCC is associated mainly with postnecrotic cirrhosis (7, 12) and that sex differences occur in the incidence of LC and in the degree of association between LC and HCC (11, 13, 18, 21, 25, 32, 33, 47, 51). We found LC twice as frequently at autopsy in males than in females, while HCC was associated with LC three times more often in males than in females during the 50-year period. It is interesting to note that a higher incidence of carcinoma in men than in women is found only in cirrhotic liver, whereas tumors in non-cirrhotic organs are equally frequent in males and females (18, 25, 32, 51). Sex differences in the incidence of LC and of LC with HCC have led to speculation on the role of gonadal hormones in these diseases, and some ex-

Table 2 Incidence of liver cirrhosis (LC) and hepatocellular carcinoma (HCC) in adults over 5-year periods, based on 22,002² autopsy reports

		Years									
		1931 -35	1936 -40	1941 -45	1946 -50	1951 -55	1956 -60	1961 -65	1966 -70	1971 -75	1976 -80
Both sexes											
Total LC	(n)	61	43	7	36	214	179	230	340	265	210
	(%)	7.7	4.1	2.2	3.7	6.3	6.0	7.2	8.6	11.0	7.4
LC with HCC	(n)	5	0	(1) ¹	3	8	10	20	28	24	19
	(%)	8.2	0	(14.3)	8.3	3.7	5.6	8.7	8.2	9.0	9.1
Males											
Total LC	(n)	38	28	6	37	133	120	157	207	162	134
% of male autopsy	(%)	7.8	4.3	2.5	4.2	6.5	6.5	8.0	8.8	11.7	8.6
LC with HCC	(n)	2	0	(1)	3	8	9	18	28	20	10
of male LC	(%)	5.3	0	(16.7)	8.1	6.0	7.5	11.5	13.5	12.4	7.5
Females											
Total LC	(n)	23	15	1	9	81	59	133	133	103	76
% of female autopsy	(%)	7.7	3.7	1.3	2.7	5.9	5.1	10.5	8.2	10.1	5.9
LC with HCC	(n)	3	0	0	0	0	1	2	0	4	9
of female LC	(%)	13	0	0	0	0	1.7	1.5	0	3.9	11.8

¹ Parenthesis indicates uncertain data due to war-time conditions.

² One hermaphrodite was excluded from 22,003 adults

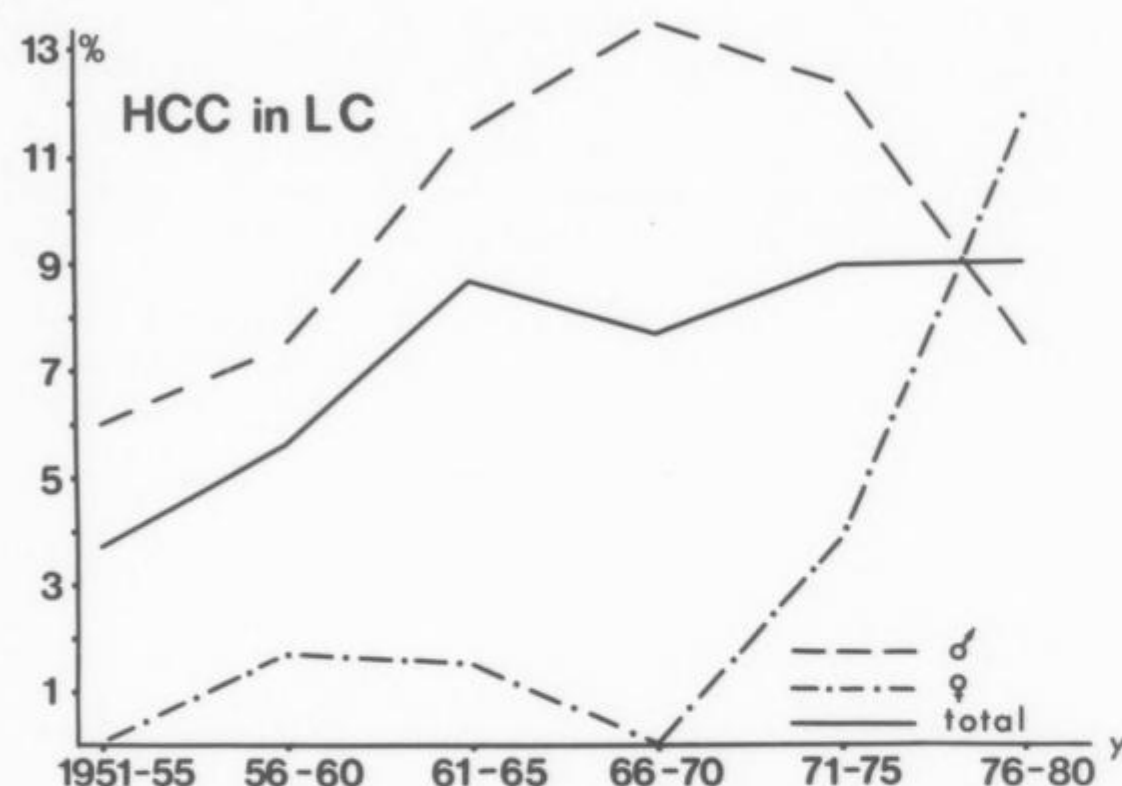


Fig. 1 Association of liver cirrhosis (LC) and hepatocellular carcinoma (HCC) in autopsy reports. Incidence in adults over a 30-year period

perimental support for such relationships has been found (14, 48, 52). However, our finding that the incidence rate of LC with HCC in males and females during the last 5 years of our study was the reverse of that seen during the previous years is difficult to explain solely on the basis of sex differences in gonadal hormones.

Evidently, additional factors play a role in the occurrence of LC and HCC in men and women.

There are several other factors that may contribute to the increased incidence of LC combined with HCC seen in most studies in recent years. One factor might be increased life-expectancy and exposure to carcinogens over a longer period (9). Another factor may be alcohol consumption (10, 33, 37, 49). A third factor may be active hepatitis-B infection. In fact, several recent epidemiological studies have shown this factor to be of possible etiological importance for the formation of primary liver cell carcinoma in patients with LC (5, 6, 15, 17, 24, 31, 35, 36, 40, 42, 50). In our opinion, the findings point to a causal connection between virus hepatitis and HCC, whereas dietary factors, alcoholism and hepatotoxins appear to be of secondary importance as cocarcinogens. Further studies are needed however, in order to determine the relative contribution of these and other factors for the increasing occurrence of HCC and LC.

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